ELECTRICAL—ELECTRIC VEHICLE (EV) RANGER COLD WEATHER BATTERY PERFORMANCE	Article No. 02-6-5

FORD: 1998-2001 RANGER

ISSUE

The EV Ranger will exhibit a reduction in the vehicle range during periods of temperatures below 21°C (70°F).

ACTION

Use the information provided as a reference for EV Ranger cool weather battery performance concerns.

SERVICE INFORMATION

EV Ranger battery range will tend to decrease when the weather turns cooler. This may be due to several factors:

- Increased lubricant friction, lower tire pressure and increased air drag will lead to higher energy consumption. For example, driving the vehicle at -1°C (30 °F) may require 40% more energy that at 21°C (70°F). This will reduce your range proportionally.
- An expected reduction in the power capability of batteries at lower temperature will reduce the time that the battery may be discharged at full power. Below 21°C (70 °F) the reduction in range is roughly one mile for every one degree Celsius (two degrees Fahrenheit) battery temperature. If the vehicle is left off-plug for extended periods, the heaters internal to the battery pack will not operate and the battery will eventually approach the external air temperature at the rate of one degree Celsius (two degrees Fahrenheit) per hour (down to the air temperature). It can thus be expected that every hour off-plug at low temperatures will result in one mile of range reduction.
- Heavy usage of the cabin Climate Control will reduce range significantly. The Climate Control heater can consume up to 6 KW; thirty minutes of continuous heater operation at this rate will reduce the range by as much as 10 miles.

These factors taken together may lead to a substantial decrease in range with modest drops in air temperature. For example, even if climate control use is minimal, expect to see a 20% (10 mile) reduction in range for an air temperature drop from 21 °C (70°F) to 4°C (40 °F) based on vehicle energy needs and reduced battery temperature. This range reduction is not permanent, the vehicle range will recover upon returning to warmer temperatures.

Several actions will help maximize range of your EV in cold weather:

- Keep the vehicle on charge whenever possible
- Drive in economy mode
- Minimize use of Climate Control
- Use the economy charge feature in order to complete charge just before use
- Charge under cover or in a garage, if possible
- Occasionally deep discharge the battery

The vehicle can be driven down to -29°C (-20 °F) with reduced range. In addition, while the battery contains heaters that are active while the vehicle is on-plug, extended periods off-plug at cold temperatures will lower the battery temperature. This may also affect the charge time, since charge of the battery will not commence until the battery is at least 4°C (40°F).

Best range is observed after a complete charge after a deep discharge. This is because the battery is heated during drive because of internal resistance and during charge because of charge inefficiencies. When the vehicle is discharged then charged back-to-back as during the typical workweek, the battery temperature rises and is maintained at a higher level. However, during weekends, typical commuters use their vehicles less often. The two days of minimal usage allows the batteries to cool down and some reduction in range might be observed after a short period of non-use.

Copyright © 2002 Ford Motor Company

The vehicle range indication on the cluster is a prediction of vehicle range based on a on-board computer which contains an algorithm that calculates the remaining vehicle range. This is a blend of past driving habits, battery state-of-charge, estimated battery capacity, and the most recent driving history. This algorithm works best when the vehicle is driven in a consistent manner and works worst when the vehicle is driven erratically.

The battery pack consists of 39 lead-acid battery modules (of approximately the size that are used in a standard internal combustion engine vehicle) connected in an electrical series. Occasionally one of these batteries fails because of wearout, or more rarely, a manufacturing defect. Since the lowest battery module limits the performance of the other 38 modules, a single defective module will limit the overall performance of the pack. Service in this situation entails replacement of the defective module. This can account for range reduction greater than expected when a ambient temperature drop is not a issue.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 203000, 203200, 290000, 614000, 614500, 614600, 622000, 690000, 698298